📘 Lecture 4 – Architectures (30 MCQs)

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1. Middleware in distributed systems can best be described as:

A) A database engine
B) The operating system of distributed systems
C) A network protocol
D) An application framework

Answer: B

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2. Middleware primarily exists to:

A) Replace operating systems
B) Implement application logic
C) Provide shared functionality for distributed applications
D) Eliminate networking

Answer: C

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3. Which of the following is NOT typically provided by middleware?

A) Communication
B) Authentication
C) Service discovery
D) User interface rendering

Answer: D

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4. A major problem when using legacy components in distributed systems is that:

A) They are too slow
B) Their interfaces are unsuitable for interoperability
C) They lack databases
D) They require modern hardware

Answer: B

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5. The main solution to legacy interface incompatibility is:

A) Rewriting the application
B) Ignoring the legacy system
C) Using a wrapper or adapter
D) Changing the operating system

Answer: C

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6. Organizing wrappers in a 1-on-1 manner for N applications requires:

A) $O(N)$ wrappers
B) $O(\log N)$ wrappers
C) $O(N^2)$ wrappers
D) $O(2N^2)$ wrappers

Answer: C

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7. Using a broker to organize wrappers reduces complexity to:

A) $O(N^2)$
B) $O(N)$
C) $O(\log N)$
D) $O(N^3)$

Answer: B

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8. In a broker-based solution, each application needs:

A) Direct interfaces to all others
B) No interfaces at all
C) One interface to send requests and one to receive responses
D) A shared database

Answer: C

9. An interceptor is best described as:

A) A database trigger
B) A hardware interrupt
C) A construct that interrupts normal control flow to execute extra code
D) A communication protocol

Answer: C

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10. The main benefit of interceptors is that they:

A) Replace middleware
B) Modify application source code
C) Allow customization without changing core middleware
D) Eliminate replication

Answer: C

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11. Request-level interceptors typically operate at:

A) The application logic level
B) The middleware invocation level
C) The operating system kernel
D) The network hardware

Answer: B

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12. Message-level interceptors are commonly used for:

A) User authentication
B) Load balancing
C) Optimizing data transfer
D) Database replication

Answer: C

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13. The basic client–server model assumes:

A) All processes are equal
B) Clients and servers follow a request–reply model
C) Servers initiate all communication
D) No separation of roles

Answer: B

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14. In a single-tier architecture:

A) Processing is distributed across layers
B) Each layer runs on a different machine
C) All processing happens on one machine
D) Clients handle only storage

Answer: C

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15. A two-tier architecture typically consists of:

A) Client and application server
B) Client and database server
C) Client and single server
D) Application server and database

Answer: C

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16. In a three-tier architecture, the middle layer is:

A) Client
B) Database server
C) Application server
D) Load balancer

Answer: C

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17. In a three-tier system, the application server acts as:

A) Only a client
B) Only a server
C) Both a client and a server
D) A passive component

Answer: C

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18. Early web servers mainly:

A) Generated dynamic content
B) Served APIs
C) Served static HTML files
D) Managed user sessions

Answer: C

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19. The introduction of CGI made web servers:

A) Simpler
B) Fully decentralized
C) Capable of dynamic content generation
D) Stateless

Answer: C

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20. Vertical distribution refers to:

A) Replicating servers horizontally
B) Splitting data among users
C) Placing logical layers on different machines
D) Using peer-to-peer communication

Answer: C

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21. Horizontal distribution refers to:

A) Separating UI, logic, and data
B) Splitting a client or server into equivalent parts

C) Centralizing control
D) Removing replication

Answer: B

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22. Peer-to-peer architectures are characterized by:

A) Dedicated servers
B) Centralized control
C) All processes being equivalent
D) Client-only nodes

Answer: C

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23. In structured P2P systems, data items are located using:

A) File paths
B) Semantic names
C) Hash-based keys
D) Broadcast queries

Answer: C

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24. A semantic-free index means:

A) Data has no meaning
B) Keys are generated independently of data semantics
C) Queries use keywords
D) Content-based search is used

Answer: B

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25. In Chord, nodes are logically organized as:

A) A tree
B) A mesh
C) A ring
D) A star

Answer: C

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26. In Chord, a data item with key k is stored at:

A) The node with the largest identifier
B) The node with identifier equal to k
C) The successor node with identifier ≥ k
D) The predecessor node of k

Answer: C

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27. The purpose of finger tables in Chord is to:

A) Store data items
B) Reduce storage overhead
C) Speed up lookup operations
D) Handle replication

Answer: C

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28. Compared to hypercube-based P2P systems, Chord is better because:

A) It requires fixed network size
B) It supports dynamic node joining
C) It uses broadcasting
D) It eliminates routing

Answer: B

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29. Hypercube-based P2P systems are less flexible because they:

A) Cannot route requests
B) Require exactly 2^d nodes
C) Use hashing
D) Are decentralized

Answer: B

30. Which architecture best fits systems like BitTorrent?

A) Client–server
B) Layered
C) Peer-to-peer
D) Service-oriented

Answer: C